

*MA.7.AR.3.2***Benchmark****MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.**

Example: Scott is mowing lawns to earn money to buy a new gaming system and knows he needs to mow 35 lawns to earn enough money. If he can mow 4 lawns in 3 hours and 45 minutes, how long will it take him to mow 35 lawns? Assume that he can mow each lawn in the same amount of time.

Example: Ashley normally runs 10-kilometer races which is about 6.2 miles. She wants to start training for a half-marathon which is 13.1 miles. How many kilometers will she run in the half-marathon? How does that compare to her normal 10K race distance?

Connecting Benchmarks/Horizontal Alignment

- MA.6.AR.3.5
- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.DP.1.3

Terms from the K-12 Glossary

- Constant of Proportionality
- Proportional Relationships

Vertical Alignment**Previous Benchmarks**

- MA.5.FR.1.1
- MA.5.AR.1.2, MA.5.AR.1.3

Next Benchmarks

- MA.7.AR.4.4, MA.7.AR.4.5
- MA.8.AR.3.1
- MA.8.GR.2.4

Purpose and Instructional Strategies

In previous courses, students developed an understanding of fraction concepts and operations, often in the context of the relationship that relates parts to wholes. In grade 6 accelerated, students solved mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system. Students will also apply ratio reasoning to solve real-world problems involving proportions. In future courses, students will determine if a linear relationship is also a proportional relationship and will solve problems involving proportional relationships between similar triangles.

- Instruction includes making connections to comparing ratios from grade 6 as a comparison using the equal sign.
 - For example, if a student can complete 7 math problems in 30 minutes and one wants to determine how many math problems they can complete in 90 minutes, they can compare the two ratios $\frac{7}{30}$ and $\frac{p}{90}$ as the equation $\frac{7}{30} = \frac{p}{90}$ to determine the number of math problems.
- Instruction does not emphasize rules, like cross multiplying, when solving proportions.
- Instruction allows time for students to analyze real-world situations. Ratio and rate reasoning can be applied to many types of real-life problems, including rate and unit rate, scaling, unit pricing, and statistical analysis (*MTR.7.1*).

Common Misconceptions or Errors

- Students may not understand the difference between an additive relationship and a multiplicative relationship. To help address this misconception, instruction includes the understanding that proportions are multiplicative relationships.
- Students may incorrectly set up proportions with one of the ratios having incorrect numbers in the numerator and denominator.
- Students may not recognize simplified forms of ratios in order to find equivalent ratios.

Strategies to Support Tiered Instruction

- Teacher provides instruction focused on the understanding of multiplicative relationships between two quantities in a proportional relationship.

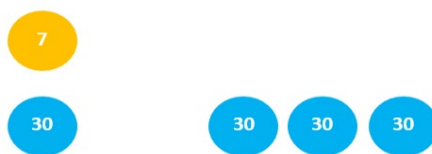
$$\frac{7}{30} = \frac{p}{90}$$

$\xrightarrow{\times 3}$
 $\xleftarrow{\times 3}$

- Teacher provides instruction on color-coding and labeling the different units when setting up a proportional relationship to ensure corresponding units are placed in the corresponding positions within the proportion.
 - For example, a student can complete **7 math problems** in **30 minutes**. How many math problems can they complete in **90 minutes**?

$$\frac{7 \text{ problems}}{30 \text{ minutes}} = \frac{x \text{ problems}}{90 \text{ minutes}}$$

- Teacher co-creates visual models with students to visualize the multiplicative relationship between quantities.
 - For example, to solve the proportion, the corresponding numbers are tripled to find a missing value of 21.



- Instruction includes the understanding that proportions are multiplicative relationships.

Instructional Tasks

Instructional Task 1 (MTR.3.1)

A recipe that makes 16 cookies calls for $\frac{1}{4}$ cup of sugar and $\frac{2}{3}$ cup of flour. Janelle wants to proportionally increase these amounts to get a new recipe using one cup of sugar.



Part A. Using the new recipe, how much flour will she need? Explain or show your work.

Part B. How many cookies can she make with the new recipe? Explain or show your work.

Instructional Task 2 (MTR.6.1, MTR.7.1)

In buying ground beef for hamburgers, there are several packages from which to choose, as shown in the table below.

Pounds of Ground Beef	1.5	2	3	3.75
Cost	\$5.58	\$7.44	\$11.16	\$13.95

If Johannes needs 5 pounds of ground beef for his barbeque, what will he pay?

Instructional Items

Instructional Item 1

Anthony is writing the place cards for his best friend's wedding reception. If he can write 12 place cards in 5 minutes, how long will it take him to complete the entire group of 180 place cards?

Instructional Item 2

Brody is working at a music store selling instruments. Brody is paid \$4 for every instrument that is sold. If he sells 6 instruments a day, how many days does he need to work to have \$120 to purchase a new pair of sneakers?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*